

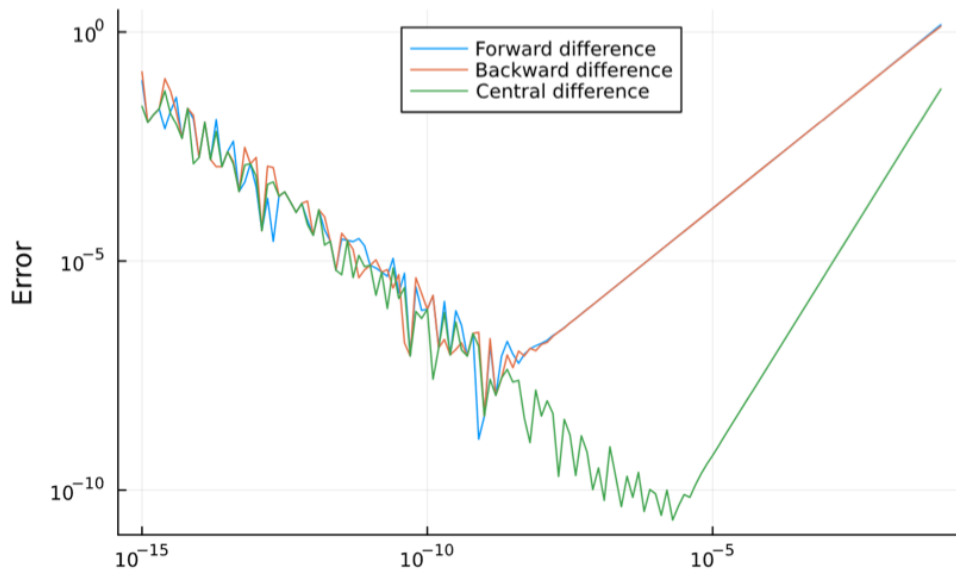
Scientific Computation

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Lecture 33 - Boundary Value Problems

Recap: Numerical Differentiation

Compare two first-order accurate methods (forward and backward difference) with a second-order accurate method (the central difference) on f' for $f(x) = \sin(x)e^x$ at $x = 0.5$:



Lecture Outline

- The boundary value problem (BVP)
- Poisson's equation in 1D
- Tridiagonal systems
- Numerical example
- Wrap-up

The Final Act

Today's topic will combine many key themes/applications of the course:

- Ordinary differential equations
- Numerical methods from Taylor series approximations
- Error analysis
- Solving linear systems of equations

We have seen each of these topics/tools in multiple contexts by now!

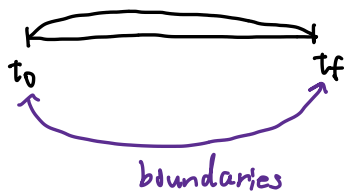
We will only solve a simple BVP but our solution strategy will capture the “essence” required to tackle much harder problems.

A reminder: BVPs are *not* on the final exam!

Boundary Value Problems

In the IVP, our solution to an ODE is fixed by an initial condition at one point. In general, we want to solve more interesting problems.

Ex. consider a vehicle whose position/velocity (i.e. state) y is governed by $\frac{dy}{dt} = f(t, y)$. We may know some initial conditions, but also have a target state at the end of some trajectory.



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Two Point BVPs: Formal Definition

We will study the special case of two point boundary value problems:

$$y'(t) = f(t, y(t)) \text{ for } y \in \mathbb{R}^n \text{ over } t \in [a, b]$$

we have $n = n_a + n_b$ boundary conditions:

$$B_a: (a, y(a)) = 0, \quad i = 1 \dots n_a$$

$$B_b: (b, y(b)) = 0, \quad i = 1 \dots n_b$$

Main difference (compare IVPs) we can't just integrate, need to consider the whole interval $[a, b]$

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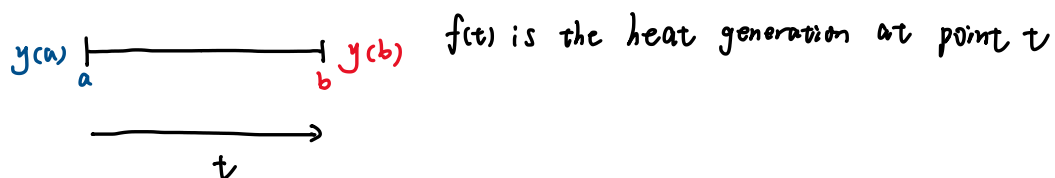
Poisson's Equation

Poisson's equation arises in multiple fields in classical mechanics including electrostatics, gravitation, and fluid dynamics. We will look at the one-dimensional version

$$y''(t) = f(t),$$

where f is called the **forcing function** and t may be a spatial dimension.

Ex. steady state temp of the 1D heat equation where $y(t)$ is the temp and t is a spatial dimension along a rod:



Poisson's Equation as a BVP

This ODE can be used in some IVPs (e.g., integrating $F = ma$), but problems like finding the steady-state temperature are two-point BVPs:

$$y''(t) = f(t), \quad y(0) = y_0, \quad y(1) = y_1,$$

where we have set $a = 0$ and $b = 1$ for simplicity.

Consider as first order:

$$y_1 = y, \quad y_2 = y_1' = y'$$

$$\Rightarrow \begin{bmatrix} y_1'(t) \\ y_2'(t) \end{bmatrix} = \begin{bmatrix} y_2(t) \\ f(t) \end{bmatrix}, \quad \left. \begin{array}{l} y_1(0) = y_0 \\ y_1(1) = y_1 \end{array} \right\} \text{boundary cond.}$$

Can't just integrate, don't know $y_2(0)$

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Solving BVPs

Just like IVPs, we will discretize our independent variable t , but we will solve for all the points at once! Consider the case where

$$a = 0$$

$$b = 1$$

$$y(0) = 0 = y(1),$$

as it will make our analysis easier. We want to satisfy $y''(t) = f(t)$ on our **grid**:

$$t_0 = a = 0 \quad t_i = h \cdot i, \quad i = 0, 1, \dots, n$$

$$t_f = b = 1 \quad h = \frac{b-a}{n}$$

$$y''(t_i) = f(t_i), \quad i = 1, \dots, n-1$$

Only consider interior grid points

→ Gives $n-1$ equations and 2 bound. cond.

→ $n-1+2 = n+1$ equations in $n+1$ unknowns y_i

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Approximating the Second Derivative

DERIVATIVE

We will use the second-order central difference formula from last class to approximate y'' : $\frac{y_0}{h^2}$

$$y''(t_i) \approx \frac{y(t_{i+1}) - 2 \cdot y(t_i) + y(t_{i-1}))}{h^2} \approx f(t_i)$$

→ This is why we only set up the ODE for interior points!

We now have $n+1$ linear equations in $n+1$ unknowns!

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Forming a Linear System

$$C_i = \frac{y_0}{h^2}$$

Gathering all of these equations gives us the following linear system:

$$-\frac{y_0}{h^2} \begin{bmatrix} -2 & 1 & 0 & \dots & 0 \\ 1 & -2 & 1 & \dots & 0 \\ 0 & 1 & -2 & 1 & \dots & 0 \\ \vdots & & & & \ddots & \\ 0 & \dots & & 1 & -2 & 1 \\ & & & 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{n-1} \end{bmatrix} = \begin{bmatrix} f(t_1) \\ f(t_2) \\ \vdots \\ f(t_{n-1}) \end{bmatrix} + \begin{bmatrix} C_1 \\ \vdots \\ C_n \end{bmatrix}$$

Where $y_0 = y(t_0) = 0 = y(t_n) = y_n$ are already accounted for
if B.C. $\neq 0$, then:

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Tridiagonal Matrices

The matrix we need to solve for our toy BVP has a **tridiagonal structure**:

$$A = \begin{bmatrix} d_1 & u_1 & & & \\ \ell_2 & d_2 & u_2 & & \\ 0 & \ell_3 & d_3 & u_3 & \\ & & \ddots & \ddots & \ddots \\ & 0 & & \ell_n & d_n \end{bmatrix} \in \mathbb{R}^{n \times n}$$

Sparse: $3n$ non zero elements out of n^2

Question: is the inverse A^{-1} sparse as well?

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Solving Tridiagonal Systems

Let's think about Gaussian elimination on a tridiagonal matrix: is it simpler?

$$\begin{bmatrix} d_1 & u_1 & & \\ d_2 & d_2 & u_2 & \\ 0 & d_3 & d_3 & u_3 \\ & 0 & & \ddots \end{bmatrix} \rightarrow \text{Each pivot only needs to clear on non zero value might below it!}$$

$\Rightarrow$ Can be solved in $O(n)$ with the Thomas algorithm. Warning: prone to numerical error without additional structure like diagonal dominance or PSD.

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Poisson's Equation Example

Example 1: Consider the two-point BVP

$$y''(t) = t - \frac{1}{2}, \quad y(0) = y(1) = 0,$$

Which has an affine forcing function $f(t) = t - 1/2$. Try $n = 2$

$$y_0 = 0 = y_2$$

$$y''(t_i) = y''(\frac{1}{2}) = 0$$

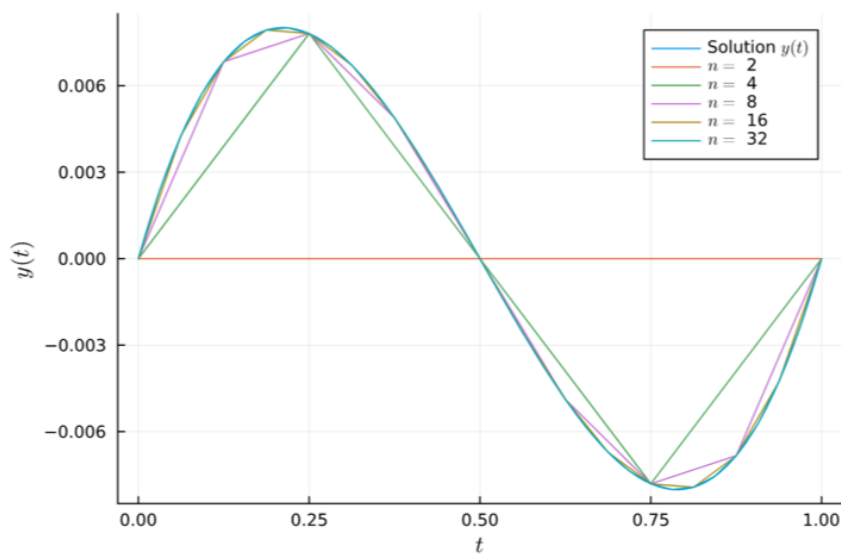
$$\Rightarrow \frac{y_0 - 2y_1 + y_2}{h^2} = 0, \quad y_1 = 0$$

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Poisson's Equation Example Cont.

Example 1 cont.: Let's compare with the actual solution

$$y(t) = \frac{1}{6}t^3 - \frac{1}{4}t^2 + \frac{1}{12}t$$



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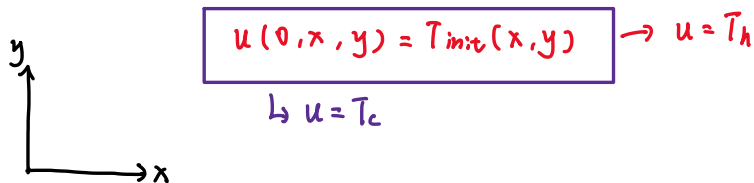
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More Advanced BVPs for Partial Differential Equations

This lecture has been an introduction to the **finite difference method** for BVPs. More exciting applications include solving the heat equation in 2D, e.g.:

$$\frac{\partial u}{\partial t} = k \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right),$$

where we specify an initial condition and boundary conditions.



$$u(0, x, y) = T_{init}(x, y) \rightarrow u = T_h$$

$\hookrightarrow u = T_c$

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One Slide Summary of the Course

1. Science and engineering mostly deal with continuous phenomena that we model with $\mathbb{R}^n$ or $\mathbb{C}^n$.
2. Computers use discrete representations of $\mathbb{R}$ (floating point numbers). This introduces **roundoff error** into calculations.
3. Nonlinear functions/equations are difficult to evaluate/solve. We use Taylor's Theorem to simplify them (typically **linearization**, i.e., first-order

approximations), introducing **truncation error**.

4. Luckily, linear systems are easy to solve and governed by the rich theory of **linear algebra**!
5. Complex problems can be solved by carefully applying this linearization strategy with an awareness of roundoff and truncation errors.

We have applied this general approach to solve systems of equations, optimization problems, and differential equations.